

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC:
INSTRUMENT CLUSTER IC [G2010733]

DESCRIPTION AND OPERATION

IPC [IPC]

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If the control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system)
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1mV or 2 K Ohm range can measure 1 Ohm. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If Diagnostic Trouble Code(s) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals
- Where an 'on demand self-test' is referred to, this can be accessed through the 'diagnostic trouble code monitor' tab on the manufacturers approved diagnostic system
- Check the JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMS which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all Diagnostic Trouble Code(s) that could be set in the instrument cluster, for additional diagnosis and testing information refer to the relevant diagnosis and testing section. For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1009-51	Ignition Authorisation - Not programed	- Instrument cluster internal failure - Target Security ID synchronisation error following reprogramming - High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the power and ground connections to the instrument cluster. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, perform Write Target Security ID routine (7006). Clear the DTC and retest - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN bus integrity test. Refer to the electrical circuit diagrams and check the CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
B1009-87	Ignition Authorisation - Missing message	- Body control module fault - CAN failure - Instrument cluster fault - Battery voltage low	- Refer to the electrical circuit diagrams and check the power and ground connections to the body control module. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN bus integrity test. Refer to the electrical circuit diagrams and check the CAN bus between the body control module and instrument cluster. Repair the wiring harness as necessary - Refer to the electrical circuit diagrams and check the power and ground connections to the instrument cluster. Repair the wiring harness as necessary - Check battery is fully charged and in serviceable condition. Refer to battery care manual
B100D-64	Column Lock Authorisation - Signal plausibility failure	Request to lock or unlock steering column lock has failed due to engine RPM or vehicle speed	Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index
B1026-12	Steering Column Lock - Circuit short circuit to battery	Steering column lock circuit short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the electric steering column ground circuit for short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
B108E-01	Display - General electrical failure	- Instrument cluster circuit short circuit to ground, short circuit to power, open circuit - Instrument cluster failure	Check the harness connector to the instrument cluster for security and integrity and check instrument cluster circuit for short circuit to ground, short circuit to power, open circuit. Repair the wiring harness as necessary, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new instrument cluster
B115C-7A	Transfer Fuel Pump - Fluid leak or seal failure	- Transfer fuel pump failure, fluid leak or seal failure - Fuel sender failure - CAN failure - Body control module failure	- Check for additional fuel pump related Diagnostic Trouble Code(s). Clear the Diagnostic Trouble Code(s) and retest - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the CAN network between the body control module and the instrument cluster. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the operation of the fuel senders. Clear the Diagnostic Trouble Code(s) and retest - If fault persists, check and install new transfer fuel pump

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1A14-96	RCM Warning Lamp - Component internal failure	- Restraints control module warning lamp LED failure - Instrument cluster failure	Check the harness connector to the instrument cluster for security and integrity. Repair the wiring harness as necessary, clear the DTC and retest. If the fault persists, install a new instrument cluster
B1A85-96	Ambient Light Sensor - Component internal failure	Internal light sensor failure	Check the harness connector to the instrument cluster for security and integrity. Repair the wiring harness as necessary, clear the DTC and retest. If the fault persists, install a new instrument cluster
P0460-11	Fuel Level Sensor A Circuit - Circuit short to ground	Active fuel level sensor circuit short circuit to ground	- Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the active fuel level sensor circuit for short circuit to ground. Repair the wiring harness as necessary
P0460-15	Fuel Level Sensor A Circuit - Circuit short to battery or open	Active fuel level sensor circuit short circuit to power, open circuit, high resistance	- Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the active fuel level sensor circuit for short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
P060A-08	Internal Control Module Monitoring Processor Performance - Bus Signal/Message Failures	Internal communication errors are causing lock-ups and resets	Check the harness connector to the instrument cluster for security and integrity. Repair the wiring harness as necessary, clear the DTC and retest. If the fault persists, install a new instrument cluster
P0610-55	Control Module Vehicle Options Error - Not configured	Car configuration file is not configured correctly	Using the Jaguar Land Rover approved diagnostic equipment check and update the car configuration file. Check engine temperature warning values are in range. Clear the DTC and retest. If the problem persists, contact the Jaguar Land Rover service desk for further assistance
P1346-11	Fuel Level Sensor B Circuit - Circuit short to ground	Passive fuel level sensor circuit short circuit to ground	- Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the passive fuel level sensor circuit for short circuit to ground. Repair the wiring harness as necessary
P1346-15	Fuel Level Sensor B Circuit - Circuit short to battery or open	Passive fuel level sensor circuit short circuit to power, open circuit, high resistance	- Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the passive fuel level sensor circuit for short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0028-82	Vehicle Communication Bus - Invalid data	- Invalid data received from another control module through the high speed CAN (powertrain) - High speed CAN (powertrain) failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (powertrain) network integrity test. Refer to the electrical circuit diagrams and check the CAN (powertrain) network. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0028-87	Vehicle Communication Bus - Missing Message	- Unexpected data has been received through the high speed CAN (powertrain) network - High speed CAN (powertrain) failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (powertrain) network integrity test. Refer to the electrical circuit diagrams and check the CAN (powertrain) network. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0028-88	Vehicle Communication Bus - Bus Off	CAN bus circuit failure (powertrain)	- Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (powertrain) network integrity test. Refer to the electrical circuit diagrams and check the CAN (powertrain) network. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0055-82	Vehicle Communication Bus - Invalid data	- Invalid data received from another control module through the high speed CAN (comfort) - High speed CAN (comfort) failure	- Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (comfort) network integrity test. Refer to the electrical circuit diagrams and check the CAN (comfort) network. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0055-87	Vehicle Communication Bus - Missing Message	■ Unexpected data has been received through the high speed CAN (comfort) network ■ High speed CAN (comfort) failure	■ Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (comfort) network integrity test. Refer to the electrical circuit diagrams and check the CAN (comfort) network. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0055-88	Vehicle Communication Bus - Bus Off	■ CAN bus circuit failure (comfort)	■ Refer to the electrical circuit diagrams and check the power and ground connections to the module. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN (comfort) network integrity test. Refer to the electrical circuit diagrams and check the CAN (comfort) network. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	■ Body control module software incompatible ■ Instrument cluster software incompatible	■ Check for other modules reporting CAN bus off or lost communication. If other modules report problems, using the Jaguar Land Rover approved diagnostic equipment check and install the latest level software to the body control module. If no other modules report problems, check and install the latest level software to the instrument cluster
U2013-02	Switch Pack - General signal failure	■ Steering wheel switchpack power or ground circuit open circuit, high resistance ■ LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Steering wheel switchpack internal failure	■ Refer to the electrical circuit diagrams and check the steering wheel switchpack power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ■ Refer to the electrical circuit diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new steering wheel switchpack
U2013-08	Switch Pack - Bus signal / message failures	■ Steering wheel switchpack LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Steering wheel switchpack left/right failure	■ Refer to the electrical circuit diagrams and check the steering wheel switchpack LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ■ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new steering wheel switchpack
U2101-56	Control Module Configuration Incompatible - Invalid / incomplete configuration	■ Unexpected car configuration data received ■ Instrument cluster is not configured correctly	■ Using the Jaguar Land Rover approved diagnostic equipment check and update the car configuration file. Clear the DTC and retest ■ If fault persists, check the harness connector to the instrument cluster for security and integrity. Reconfigure the instrument cluster using the Jaguar Land Rover approved diagnostic equipment. Clear the Diagnostic Trouble Code(s) and retest. If fault persists, install a new instrument cluster

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-46	Control Module - Calibration / parameter memory failure	Instrument cluster internal failure - Odometer storage corrupted	Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest. If fault persists, install a new instrument cluster
U3000-55	Control Module - Not configured	- Incorrect wheels/tyres installed - Tire size compensation is not configured correctly	- Install the correct wheels/tyres - Using the Jaguar Land Rover approved diagnostic equipment check and update the car configuration file. Clear the DTC and retest
U3002-81	Vehicle Identification Number - Invalid serial data received	Invalid Vehicle Identification Number	Using the Jaguar Land Rover approved diagnostic equipment check and update the car configuration file. Clear the DTC and retest
U3003-17	Control Module - Circuit voltage above threshold	Circuit voltage is above threshold	Check vehicle battery and charging system. Refer to the relevant section in the workshop manual. Refer to the electrical circuit diagrams and check the power supply circuits to the module. Clear the Diagnostic Trouble Code(s) and retest